# Economics S5040: Dynamic Stochastic Modeling
# Lecture 12 — Calibrating the Solow Model to U.S. Data

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## 1 The Question for Today

Last time we derived

$$k_{t+1} = sAk_t^{\alpha} + (1-\delta)k_t.$$

Today:

| Can this simple model describe the U.S. economy quantitatively? |
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To answer, we connect model objects to measured objects.

## 2 What Is Calibration?

A model contains parameters

$$\theta = (\alpha, s, \delta, A, \ldots).$$

Calibration chooses numerical values using facts from the economy we want to study:

| model → empirical counterparts → parameters → predictions. |
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A useful model will generally match some facts and miss others. Those misses are informative.

## 3 The Penn World Table

The Penn World Table measures output, inputs, and productivity across countries and over time. PWT 11.0 covers 185 countries through 2023.

We focus on the United States. PWT contains several concepts of real GDP and capital. They are not interchangeable. For studying one country's evolution through time, we want appropriate national-accounts time-series concepts rather than blindly choosing a PPP level series.

| Always read the definition of a PWT variable before using it. |
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## 4 Model Objects and Data Objects

Our model is

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}.$$

We need counterparts for $Y_t, K_t, L_t, \alpha, s, \delta$.

Useful PWT fields for a U.S. time-series exercise include

| Model object | Data concept | PWT field |
|---|---|---|
| $Y_t$ | real GDP, national-accounts series | `rgdpna` |
| $K_t$ | real capital stock, national-accounts series | `rnna` |
| $L_t$ | persons engaged | `emp` |
| $1-\alpha$ | labor share | `labsh` |
| $\delta_t$ | depreciation rate | `delta` |
| $s_t$ | investment/output counterpart | investment expenditure data |

We will check these against the PWT 11.0 codebook before constructing the final student notebook.

## 5 Step 1: Inspect the U.S. Data

A typical workflow is

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

pwt = pd.read_excel("pwt110.xlsx", sheet_name="Data")
us = pwt[pwt["countrycode"] == "USA"].copy()
us = us.sort_values("year")

us[["year", "rgdpna", "rnna", "emp",
    "labsh", "delta"]].tail()
```

Before calculating anything, ask what each series measures and what its units are.

## 6 Step 2: Calibrate the Capital Share

Competitive Cobb–Douglas production implies

$$wL = (1-\alpha)Y.$$

Therefore

$$\boxed{\alpha_t = 1 - \texttt{labsh}_t.}$$

For a simple calibration, use a sample average:

$$\boxed{\alpha = 1 - \frac{1}{T}\sum_t \texttt{labsh}_t.}$$

The simple model treats $\alpha$ as a parameter, not as a new parameter every year.

## 7 Step 3: Capital and Output per Worker

Construct

$$\boxed{k_t = \frac{K_t}{L_t}}, \qquad \boxed{y_t = \frac{Y_t}{L_t}}.$$

Plot both through time. Solow predicts that capital deepening raises output per worker. This gives us our first visual confrontation of model and data.

## 8 Step 4: The Capital–Output Ratio

Construct

$$\frac{K_t}{Y_t}.$$

At a steady state with no population or productivity growth,

$$sY^* = \delta K^*,$$

so

$$\boxed{\frac{K^*}{Y^*} = \frac{s}{\delta}.}$$

We can therefore compare the observed $K/Y$ ratio with the model-implied $s/\delta$.

## 9 Step 5: Depreciation

One approach is to use an average of PWT's depreciation-rate measure:

$$\delta = \text{average}(\delta_t).$$

But the accumulation equation also tells us how depreciation can be inferred:

$$K_{t+1} = (1 - \delta_t)K_t + I_t.$$

Hence

$$\boxed{\delta_t = \frac{I_t + K_t - K_{t+1}}{K_t}.}$$

Even if we use PWT's measure, this derivation connects the parameter to economic accounting.

## 10 Step 6: The Saving Rate

In Solow,

$$I_t = sY_t,$$

so the natural empirical counterpart is

$$\boxed{s_t = \frac{I_t}{Y_t}.}$$

For a benchmark calibration,

$$s = \frac{1}{T}\sum_t s_t.$$

Actual U.S. saving and investment need not coincide because the United States is an open economy. The investment share is nevertheless the natural counterpart to the flow $sY$ that creates capital in the Solow accumulation equation.

## 11 Step 7: Construct Productivity

From

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha},$$

we obtain

$$\boxed{A_t = \frac{Y_t}{K_t^{\alpha} L_t^{1-\alpha}}.}$$

Equivalently,

$$\boxed{A_t = \frac{y_t}{k_t^{\alpha}}.}$$

This is a residual. Anything affecting measured output that is not captured by measured capital and labor appears in $A_t$. Hence measured TFP should not automatically be interpreted as pure technology.

## 12 A First Growth-Accounting Equation

Taking logs,

$$\log Y_t = \log A_t + \alpha \log K_t + (1-\alpha) \log L_t.$$

Taking changes gives approximately

$$\boxed{\Delta \log Y_t = \Delta \log A_t + \alpha \Delta \log K_t + (1-\alpha) \Delta \log L_t.}$$

Output growth is decomposed into productivity, capital, and labor growth.

## 13 Calibrating a Benchmark U.S. Economy

Choose a sample period and calculate

$$\hat{\alpha}, \qquad \hat{s}, \qquad \hat{\delta}, \qquad \hat{A}.$$

Then

$$k_{t+1} = \hat{s}\hat{A}k_t^{\hat{\alpha}} + (1-\hat{\delta})k_t,$$

with steady state

$$\boxed{\hat{k}^* = \left(\frac{\hat{s}\hat{A}}{\hat{\delta}}\right)^{1/(1-\hat{\alpha})}.}$$

Compare $\hat{k}^*$ with observed U.S. capital per worker.

## 14 Is the United States at Its Solow Steady State?

Starting from observed $k_0$, simulate

$$k_1 = g(k_0), \quad k_2 = g(k_1), \quad \ldots$$

If $k_0 < \hat{k}^*$, the calibrated model predicts capital deepening. If $k_0 > \hat{k}^*$, it predicts declining capital per worker.

The question is now empirical:

$$\boxed{\text{Given U.S. parameters, where does Solow say the economy is headed?}}$$

## 15 Successive Approximation

The simplest computation iterates the law of motion:

$$k_{n+1} = g(k_n).$$

```
def g(k, A, s, alpha, delta):
    return s * A * k**alpha + (1-delta) * k

k = k0
for n in range(100):
    k = g(k, A, s, alpha, delta)
```

This computation has an economic interpretation: it follows the model economy through time.

## 16 The Steady State as a Root

A fixed point satisfies

$$g(k^*) = k^*.$$

Define

$$h(k) = g(k) - k = sAk^\alpha - \delta k.$$

Then

$$\boxed{h(k^*) = 0.}$$

The Cobb–Douglas case has a closed form, but many economic models do not.

## 17 Newton's Method

To solve $h(k) = 0$, start from $k_n$ and use the zero of the tangent line:

$$\boxed{k_{n+1} = k_n - \frac{h(k_n)}{h'(k_n)}.}$$

Here

$$h'(k) = \alpha sAk^{\alpha-1} - \delta.$$

Therefore

$$k_{n+1} = k_n - \frac{sAk_n^\alpha - \delta k_n}{\alpha sAk_n^{\alpha-1} - \delta}.$$

Newton's method generally approaches the fixed point much faster than simply iterating the law of motion.

But distinguish the two ideas:

- successive approximation follows economic dynamics;
- Newton's method searches directly for the fixed point.

## 18 A Model–Data Diagnostic

Compute

$$\left(\frac{K}{Y}\right)_{\text{data}} \qquad \text{and} \qquad \frac{\hat{s}}{\hat{\delta}}.$$

If they differ, possible reasons include:

- the United States is not at a steady state;
- population and productivity are growing;
- investment and capital concepts do not align perfectly;
- depreciation varies;
- the one-capital-good model is too simple.

A discrepancy tells us what the simple model leaves out.

## 19 Productivity Growth and Balanced Growth

The introductory Solow model initially holds $A$ fixed. U.S. productivity does not remain fixed forever. If $A_t$ grows, the target toward which capital moves also changes.

Thus actual U.S. output per worker need not converge to a constant:

$$\boxed{\text{steady state in levels} \quad \text{versus} \quad \text{balanced growth.}}$$

We do not need the full balanced-growth model yet. The distinction explains why a fixed-$A$ model cannot literally reproduce long-run U.S. growth in output per worker.

## 20 What Did Calibration Buy Us?

Before calibration:

$$s \uparrow \Rightarrow k^* \uparrow .$$

After calibration we can ask how large U.S. $\alpha, s, \delta$ actually are, what capital per worker the model predicts, how far the United States is from that benchmark, and how much measured growth is associated with capital, labor, and the productivity residual.

The model now makes numerical statements about an actual economy.

## 21 The Question Solow Cannot Answer

We calibrated $s$ from U.S. data, but where did $s$ come from?

$$\boxed{s = \text{given.}}$$

Solow does not explain why households save that amount.

That is the transition to Week 7: instead of assuming saving, households will choose consumption and saving optimally.

## 22 What to Remember

1. Calibration connects parameters to measured quantities.
2. PWT provides empirical counterparts for output, capital, labor, labor shares, depreciation, investment, and productivity.
3. PWT variable definitions matter.
4. Under Cobb–Douglas, $\alpha = 1 -$ labor share.
5. The Solow residual is $A_t = Y_t/(K_t^{\alpha} L_t^{1-\alpha})$.
6. In the simple steady state, $K^*/Y^* = s/\delta$.
7. Successive approximation follows the economic dynamics.
8. Newton's method searches directly for the fixed point.
9. Calibration lets us confront Solow with U.S. data.
10. Solow leaves the saving rate unexplained; Week 7 will make it a choice.